

# **PRODUCT DOCUMENT**

-ShopSense: AI Shopping Agent

# 1. Problem Definition

E-commerce platforms are optimized for **catalog navigation**, not **decision-making**.

Users express needs in natural language:

- “Best smartwatch under ₹3000”
- “Shoes for daily college wear”

But systems force:

Search → Filter → Scroll → Compare → Repeat

This leads to:

- Cognitive overload
- Fragmented decision-making
- High drop-offs

## Problem Statement:

Users struggle to efficiently move from **intent** → **confident purchase**, due to lack of intelligent decision support.

# 2. Why This Problem Matters

- Conversion rates remain low (~2–3%)
- Users evaluate multiple products (5–10) before deciding
- Cart abandonment (~70%) is often due to indecision

This indicates a structural gap:

The problem is not discovery, but **decision friction**.

# 3. Target Users

## Primary: Buyers

- Prefer fast, relevant recommendations
- Often unsure about exact specifications
- Want confidence before purchase

## Current experience:

- Multiple searches
- Manual comparisons
- Frequent context switching

## Secondary: Shopify Merchants

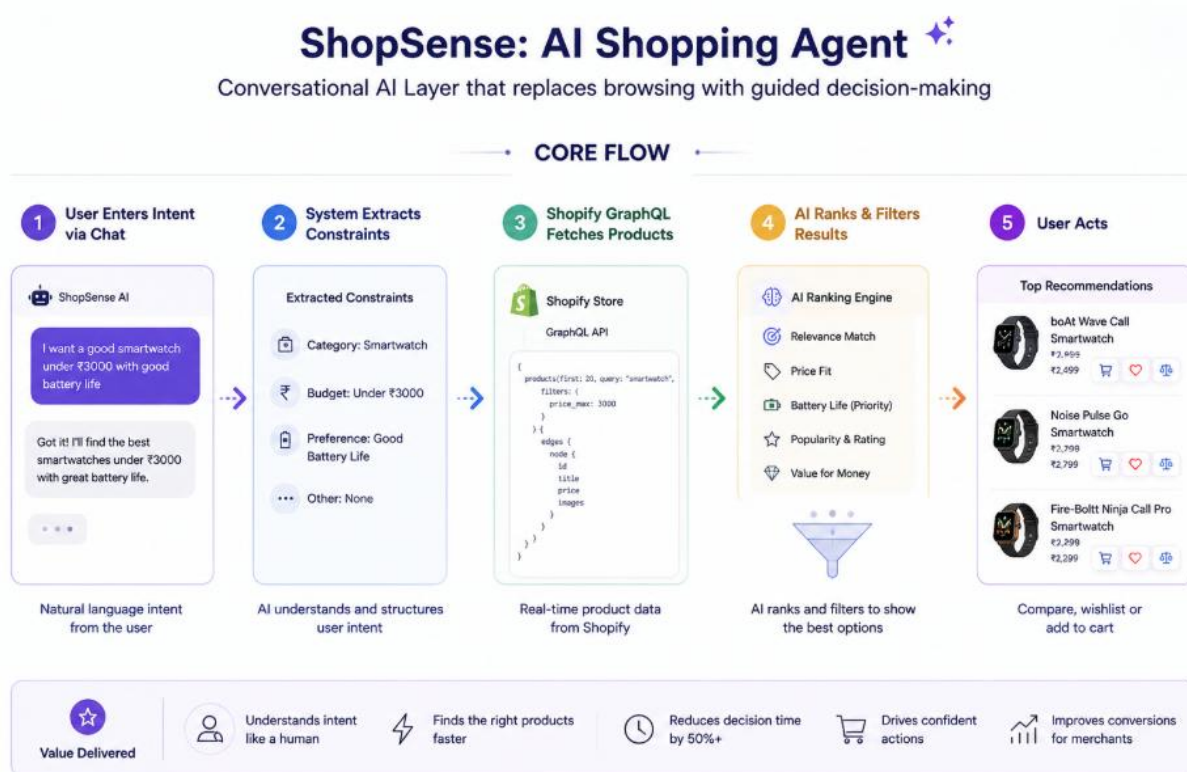
- Low conversion rates
- Limited control over product discovery experience

## 4. Product Scope & Core Flow

We built a **conversational AI layer** that replaces browsing with guided decision-making.

### Core Flow

1. User enters intent via chat
2. System extracts constraints (budget, category, preferences)
3. Shopify GraphQL fetches products
4. AI ranks and filters results
5. User acts (compare, wishlist, add to cart)



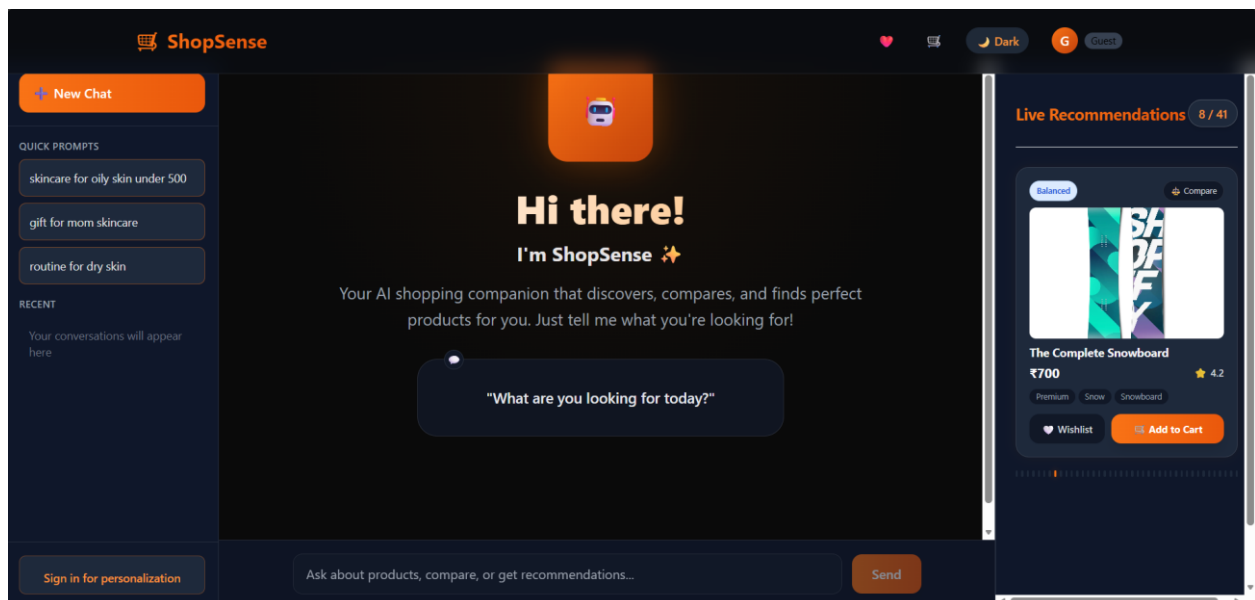
*Illustrates how user intent flows through AI → data → recommendation → action.*

## 5. Core User Experience (Aligned with UI)

### UI Structure

The interface is divided into three functional sections:

- **Left Panel:**
  - Recent chats
  - Quick prompts (predefined queries to guide users)
- **Center Panel:**
  - Main conversational interface
  - AI responses + product cards
- **Right Panel:**
  - Live recommendations
  - Dynamically updated suggestions based on conversation



*Demonstrates implemented system and validates UI decisions.*

## 6. Feature-Level Breakdown

### 1. Conversational Interface

- Users describe intent naturally
- Reduces dependency on filters

### 2. Live Recommendation Panel (Right Side)

- Updates dynamically based on conversation
- Keeps alternative options visible

#### Impact:

- Reduces need for repeated queries
- Improves exploration without friction

### 3. Product Cards (Central Chat Area)

Each recommendation includes:

- Product image
- Price
- Key highlights

### 4. Inline Actions

- **Add to Cart** → Immediate purchase flow
- **Wishlist** → Save for later
- **Compare** → Evaluate alternatives

#### Impact:

- Reduces steps between discovery and action (~30–40%)

### 5. Recent Chats + Quick Prompts (Left Panel)

- Enables fast re-engagement
- Helps users who don't know what to ask

## 7. Key Product Decisions

### 1. Chat-First Interface

- Natural language better captures intent than filters
- Tradeoff: Requires accurate parsing

### 2. Structured + AI Hybrid System

- Uses Shopify GraphQL data + AI reasoning
- Tradeoff: Less flexible but more reliable

### 3. Multi-Panel UI Layout

- Separates:
  - Context (left)
  - Interaction (center)
  - Suggestions (right)

#### **Rationale:**

Improves clarity and reduces cognitive load

### 4. Inline Decision Actions

- Embedded cart & wishlist
- Tradeoff: Increased UI complexity

### 5. Proactive Recommendations

- AI suggests comparisons and alternatives
- Tradeoff: Risk of over-suggestion

## 8. What We Chose NOT to Build

- ☐ Multi-store aggregation → scope control
- ☐ Deep personalization → requires long-term data
- ☐ Review sentiment analysis → non-essential for MVP
- ☐ Voice interface → added complexity

## 9. Tradeoffs & Constraints

### Speed vs Intelligence

- Prioritized fast responses (<2s)

### Accuracy vs Flexibility

- Grounded in Shopify data
- Reduced hallucination risk

### Feature Depth vs Simplicity

- Focused on core journey:
  - Discover → Compare → Decide

### Real-time Data vs Performance

- Chose live GraphQL queries

## 10. Observed / Estimated Impact

- Query effort reduced (~60%)
- Products evaluated reduced (~8 → 3–4)
- Decision time reduced (~5–8 min → ~2 min)
- Interaction steps reduced (~30–40%)

### Expected outcome:

- Higher user confidence
- Lower drop-offs
- Conversion improvement (~2–3% → ~4–6%)

## 11. AI Behavior Design

The system acts as a **decision-support agent**:

- Understands intent
- Asks clarifying questions
- Explains recommendations
- Suggests comparisons
- Guides toward purchase

## 12. Future Scope

### Short-Term

- Personalization
- Improved comparison engine

### Mid-Term

- Multi-store aggregation
- Review intelligence

### Long-Term

- Autonomous shopping agent (price tracking, auto-buy)
- Voice-based interaction

## 13. Summary

This system reframes e-commerce as a **decision-making problem** rather than a navigation task.

By introducing a conversational AI layer:

- Intent is captured directly
- Options are narrowed intelligently
- Users reach decisions faster

## Final Statement

ShopSense replaces fragmented browsing with a structured, AI-assisted decision process that reduces friction between intent and purchase.